No. of Functional Features Included in the Solution

SMARTBRIDGE - No. of Functional Features Included in the Solution

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-7817
- **Team Size:** 4
- **Team Leader:** Karan S
- **Team Members:** SANJAY .M, Santhosh S, Yaseen P
- **Project Name:** AI FitTrack
- **Maximum Marks:** 5 Marks

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	**User Registration**	Creates a new user account with encrypted bcrypt password hashing.	Auth Module (authController.js)	**Done**	0.5 Marks
2	**User Login & JWT Auth**	Verifies credentials and generates a signed JWT token.	Auth Module (authController.js)	**Done**	0.5 Marks
3	**Get User Profile**	Fetches profile details for the authenticated bearer token.	Auth Module (authController.js)	**Done**	0.5 Marks

4	**Add Workout Record**	Logs a workout with name, category, duration, calories, and date.	Workout Module (workoutController.js)	**Done**	0.5 Marks
5	**Get / Search Workouts**	Retrieves all workouts or filters history by workout name, category, or date.	Workout Module (workoutController.js)	**Done**	0.5 Marks
6	**Update / Delete Workout**	Modifies or deletes an existing workout record linked to the user.	Workout Module (workoutController.js)	**Done**	0.5 Marks
7	**AI Workout Plan Generator**	Queries Gemini AI (@google/genai) using age, goals, and experience level.	AI Module (aiController.js, geminiService.js)	**Done**	1.0 Marks
8	**AI Fitness Analytics Insights**	Analyzes total workouts, avg duration, and calories burned to return insights.	AI Module (aiController.js, geminiService.js)	**Done**	1.0 Marks

Feature Summary

Metric	Count / Value
Total Features Planned	**8**
Total Features Implemented	**8**
Core / Must-Have Features	**6**
Additional / Nice-to-Have Features	**2**
Features Tested & Verified	**8 (100%)**

Feature Category Breakdown

S.No	Category PDF	Features in Category PDF	Example Features PDF
1	**User Interface (UI)**	1	REST API JSON Consumer Interface / Postman / Thunder Client
2	**Backend / Logic**	3	Express Routing, Controller Business Logic, Error Handler Middleware
3	**Database / Storage**	2	MongoDB Atlas Users Collection, Workouts Collection

4	**API / Integration**	2	Google Gemini AI Recommendation & Fitness Insight Service Integration
5	**Security / Authentication**	2	JWT Bearer Authentication, bcryptjs Password Hashing